

Ben Clingenpeel

Washington, DC
Email: ben.clingenpeel@gwu.edu

Education

2023-present	George Washington University PhD in Mathematics
2019-2023	Georgetown University Bachelor of Science in Mathematics

Interests

I am a third year PhD student at George Washington University and am interested in pursuing a career in mathematical research and teaching at the college level. I am interested in knot theory and my advisor is Professor Józef H. Przytycki.

Papers

2025	In search of homology for quasigroups of Bol-Moufang type. With Anthony Christiana, Huizheng (Ali) Guo, Jinseok Oh, Józef H. Przytycki, and Anna Zamojska-Dziena. arXiv:2508.21268 .
2025	Low dimensional homology of the Yang-Baxter operators yielding the HOM-FLYPT polynomial. With Anthony Christiana, Huizheng (Ali) Guo, Jinseok Oh, Józef H. Przytycki, Xiao Wang, and Hongdae Yun. arXiv:2502.20659 .
2022	Invariants of partial knots and symmetric union presentations. With Zongzheng Dai, Gabriel Diraviam, Kareem Jaber, Krishnendu Kar, Ziyun Liu, Teo Micklethun, Haritha Nagampoozhy, Michael Perry, Moses Samuelson-Lynn, Eli Seamans, Ana Wright, Nicole Xie, Ruiqi Zou, and Alexander Zupan. arXiv:2504.08589 .

Presentations

November 2025	Symmetric unions of knots. Talk given at the George Washington University Graduate Student Seminar.
October 2025	Computing homology of the Yang-Baxter operators yielding the HOMFLYPT polynomial. Talk given at the upcoming Special Session on Inverse Problems of the AMS Fall Central Sectional Meeting in St. Louis, MO.

October 2025	Homology of Yang-Baxter Operators. Specialty exam talk given at George Washington University.
May 2025	Low dimensional homology of the Yang-Baxter operators yielding the HOM-FLYPT polynomial. Given at the Special Session on Yang-Baxter operators in Knot Theory of the AMS Spring Western Sectional Meeting in San Luis Obispo, CA.
April 2025	Low dimensional homology of the Yang-Baxter operators yielding the HOM-FLYPT polynomial. Distinguished Graduate Student talk given at <i>Knots in Washington 51</i> at the George Washington University in Washington, DC.
January 2023	Colorings of knots with symmetric union presentations. Talk given with Eli Seamans at the special session of the Joint Mathematics Meeting on the Polymath Jr program in Boston, MA.

Teaching and Tutoring

July 2024 - Present	Instructor of Record George Washington University • Taught MATH 1231, Calculus I (Summer 2024). • Taught MATH 1232, Calculus II (Summer 2025).
Aug 2023 - Present	Graduate Teaching Assistant George Washington University • Graded and taught recitation sections for MATH 1231, Calculus I (Fall 2023, Spring 2024, Fall 2024). • Graded and taught recitation sections for MATH 1232, Calculus II (Spring 2025). • Graded and taught recitation sections for MATH 2184, Linear Algebra I (Fall 2025, Spring 2026). • Tutored students in lower division math courses through the Calculus Lab (all semesters).
Jan 2022 - May 2023	Math Tutor Georgetown University Math Assistance Center • Tutored Georgetown University students in lower division courses. • Led trainings for new tutors.
Aug 2021 - Dec 2022	Math Teaching Assistant and Grader Georgetown University • Graded student assignments in MATH 215, Abstract Algebra (Spring 2022, Fall 2022). • Graded student assignments in MATH 150, Linear Algebra and held a weekly office hour (Fall 2021).

Jan 2020 - **Math Tutor** | Georgetown University GUMSHOE Program
 May 2020

- Volunteered with a campus math and science tutoring group.
- Worked with middle school students from DCs seventh and eighth wards who came to campus for extra weekend math lessons.

Sep 2017 - **Math Tutor** | Freelance
 Present

- Tutored students at a variety of different levels, some in middle school and high school classes, as well as a few in first college calculus courses.
- Designed curricula for students who weren't following a given course schedule. For these students, I selected appropriate topics in math that don't typically appear in middle school and high school curricula.

Conference Activities

Name	Location	Role(s)
Knots in Washington 52 (December 5-7, 2025)	Washington, DC (George Washington University)	Co-organizer
AMS 2025 Fall Central Sectional Meeting	St. Louis, MO (Saint Louis University)	Speaker: <i>Computing homology of Yang-Baxter operators yielding the HOMFLYPT polynomial</i>
AMS 2025 Spring Western Sectional Meeting	San Luis Obispo, CA (California Polytechnic State University)	Speaker: <i>Low dimensional homology of Yang-Baxter operators yielding the HOMFLYPT polynomial</i>
Knots in Washington 51 (April 4-6, 2025)	Washington, DC (George Washington University)	Distinguished Graduate Student Speaker: <i>Low dimensional homology of Yang-Baxter operators yielding the HOMFLYPT polynomial</i> Co-organizer
Knots in Washington 50 (December 6-8, 2025)	Washington, DC (George Washington University)	Co-organizer

Knots in Washington 49.984375 (April 26-28, 2024)	Washington, DC (George Washington University)	Co-organizer
AMS 2024 Spring Eastern Sectional Meeting	Washington, DC (Howard University)	Attendee
Knots in Washington 49.96875 (December 8-10, 2023)	Washington, DC (George Washington University)	Co-organizer
Joint Mathematics Meetings 2023	Boston, MA	Speaker: <i>Colorings of knots with symmetric union presentations</i>

Honors and Awards

- 2025 | **Distinguished Graduate Student Speaker.** *Knots in Washington 51.*
- 2023 | **Henry M. Leslie Award.** Awarded to that member of the senior class showing highest proficiency in Mathematics. *Georgetown University.*